Where irrigation exists is globally contested

5. Analyses supporting the reply to the reviewers

Arnald Puy

Contents

1 Preliminary

```r
# ANALYSES SUPPORTING THE REPLY TO THE REVIEWERS ##############################
###############################################################################

# This script reproduces every quantitative figure cited in our reply to the
# reviewers. All analyses use the harmonized ensemble at 0.2 deg restricted to
# cropland cells (the same universe as the main analysis).

# Load libraries --------------------------------------------------------------

sensobol::load_packages(c("data.table", "here", "terra", "magrittr"))

# Functions reused from the main analysis --------------------------------------

source(here("functions", "filter_long_dt_by_crop_mask_aggregated.R"))
source(here("functions", "compute_tau_max_disagreement_fun.R"))

# Harmonized ensemble at 0.2 deg, restricted to cropland -----------------------

dt <- fread(here("datasets", "irrigated_areas_regridded",
                 "irrigated_areas_regridded_02.csv"))
dt[, resolution := "0.2deg"]
crop <- rast(here("irrigated_area_datasets", "GirrEO", "asap_mask_crop_v02.tif"))
dt <- filter_long_dt_by_crop_mask_aggregated(dt = dt, crop_native = crop,
                                             rule = "any")

# Cells x maps -----------------------------------------------------------------

wide <- dcast(dt, lon + lat + country + continent ~ dataset, value.var = "mha",
              fill = 0)

maps <- c("gaez_v4", "giam", "gmia", "gripc", "luh2", "meier",
          "mirca2000", "mirca_os", "nagaraj", "spam")
stopifnot(all(maps %in% names(wide)))

# Nominal reference year of each map (Table on temporal characteristics) -------

nominal <- c(giam = 2000, mirca2000 = 2000, gmia = 2005, gripc = 2005,
             nagaraj = 2005, meier = 2005, spam = 2010, mirca_os = 2015,
             gaez_v4 = 2015, luh2 = 2015)

# Mapping families -------------------------------------------------------------

census <- c("gmia", "mirca2000", "mirca_os", "spam", "gaez_v4")  # FAO-statistics
rs      <- c("giam", "gripc")                                    # remote sensing
```

```r
# Detectability threshold grid (1 ha to 1e5 ha) and 1%-of-cell tolerance -------

taus_ha     <- c(0, 10^(seq(0, 5, by = 0.1)))
taus_mha    <- taus_ha * 1e-6
zero_tol_02 <- 500  # 1% of the 0.2 deg cell, in ha

# Existential disagreement: of the cells where at least one map records
# irrigation (> 0), the fraction in which at least one map records exactly zero.

exist_union <- function(w, cols) {
  A <- as.matrix(w[, ..cols])
  n_present <- rowSums(A > 0, na.rm = TRUE)
  disagree  <- n_present > 0 & n_present < length(cols)
  Amin <- do.call(pmin, c(as.data.frame(A), list(na.rm = TRUE)))
  sum(disagree & Amin == 0) / sum(n_present > 0)
}

# Extreme disagreement: fraction of cells whose tau_max exceeds ten times the
# detectability threshold (one map reports > ~10% of the cell, another ~0).

extreme_frac <- function(w, cols, zero_tol = zero_tol_02) {
  tm <- compute_tau_max_existential_fun(w, cols, taus_mha, taus_ha,
                                         zero_tol_ha = zero_tol)
  mean(!is.na(tm$tau_max_ha) & tm$tau_max_ha > 10 * zero_tol)
}
```

2 Reviewer 1

2.1 R1.1: the detectability threshold as a fraction of the grid cell

The same absolute τ (in hectares) is a different fraction of the grid cell at each resolution. We report the equator cell areas and translate key τ values into cell fractions.

```r
# Equator cell area (ha) per resolution ----------------------------------------

R_earth <- 6371
deg <- c("0.2" = 0.2, "0.4" = 0.4, "1" = 1)
cell_ha <- sapply(deg, function(d) (d * pi / 180 * R_earth)^2 * 100)
round(cell_ha)
```

```
##     0.2     0.4       1
##   49457  197829 1236431
```

```r
# Key tau values as a percentage of the cell ----------------------------------

tau_key <- c(100, 500, 1000, 2500, 5000, 10000, 25000)
tau_tab <- data.table(
  `tau (ha)`     = tau_key,
  `0.2 deg (%)` = round(100 * tau_key / cell_ha["0.2"], 2),
  `0.4 deg (%)` = round(100 * tau_key / cell_ha["0.4"], 2),
  `1 deg (%)`    = round(100 * tau_key / cell_ha["1"],   2))

knitr::kable(tau_tab, caption = "Detectability threshold tau as a percentage of the grid-cell a
```

Table 1: Detectability threshold tau as a percentage of the grid-cell area at the equator. The analysis centres on tau = 1% of the cell (about 500, 2000 and 12400 ha); extreme disagreement denotes tau_max above ten times this value.

tau (ha)	0.2 deg (%)	0.4 deg (%)	1 deg (%)
100	0.20	0.05	0.01
500	1.01	0.25	0.04
1000	2.02	0.51	0.08
2500	5.05	1.26	0.20
5000	10.11	2.53	0.40
10000	20.22	5.05	0.81
25000	50.55	12.64	2.02

2.2 R1.5: temporal heterogeneity does not explain the disagreement

We test whether the disagreement reflects the different moments the maps represent. Existence of irrigation is the metric throughout (see Preliminary).

```r
# Maps that share the 2005 reference year versus the full ensemble --------------

sameyear_2005 <- c("gmia", "gripc", "nagaraj", "meier")

data.table(
```

```r
  set  = c("Same year (2005)", "Full ensemble"),
  maps = c(length(sameyear_2005), length(maps)),
  existential_pct = round(100 * c(exist_union(wide, sameyear_2005),
                                  exist_union(wide, maps))))
```

```
##                 set  maps existential_pct
##              <char> <int>           <num>
## 1: Same year (2005)     4              69
## 2:    Full ensemble    10              90
```

```r
# Recompute over every four-map combination to hold ensemble size fixed --------

combs4  <- combn(maps, 4, simplify = FALSE)
ev4     <- sapply(combs4, function(cc) exist_union(wide, cc))
span4   <- sapply(combs4, function(cc) diff(range(nominal[cc])))
same_ev <- exist_union(wide, sameyear_2005)

data.table(
  quantity = c("Minimum", "Median", "Maximum",
               "Same-year (2005) combination", "Most dispersed (1999-2015)"),
  existential_pct = round(100 * c(min(ev4), median(ev4), max(ev4),
                                  same_ev, mean(ev4[span4 == max(span4)]))),
  percentile = c(NA, NA, NA, round(100 * mean(ev4 <= same_ev)), NA))
```

```
##                        quantity existential_pct percentile
##                          <char>           <num>      <num>
## 1:                      Minimum              15         NA
## 2:                       Median              74         NA
## 3:                      Maximum              89         NA
## 4: Same-year (2005) combination              69         36
## 5:   Most dispersed (1999-2015)              74         NA
```

```r
# Direction of disagreement: under real expansion, a recent (2015) map should
# not record zero where an older (2000-2005) map detects irrigation.

old_t    <- c("giam", "mirca2000", "gmia", "gripc")    # 2000-2005
recent_t <- c("gaez_v4", "luh2", "mirca_os")           # 2015

pooled_blind <- function(new) {
  A_old   <- as.matrix(wide[, ..old_t])
  old_any <- rowSums(A_old > 0, na.rm = TRUE) > 0
  sum(old_any & wide[[new]] == 0, na.rm = TRUE) / sum(old_any)
}

data.table(
  recent_map = c("GAEZ+2015", "LUH2", "MIRCA-OS"),
  sees_zero_where_older_detects_pct = round(100 * sapply(recent_t, pooled_blind)))
```

```
##    recent_map sees_zero_where_older_detects_pct
```

```
##        <char>                                   <num>
## 1:  GAEZ+2015                                     19
## 2:       LUH2                                     83
## 3:   MIRCA-OS                                     17
```

3 Reviewer 2

3.1 R2.1: census products agree because they share source data, not reality

The five FAO-statistics products share definitions and much of their source data. We compare their existential disagreement against all five-map subsets.

```
census_ev <- exist_union(wide, census)

combs5 <- combn(maps, 5, simplify = FALSE)
ev5    <- sapply(combs5, function(cc) exist_union(wide, cc))

data.table(
  quantity = c("Census-5 (GMIA, MIRCA2000, MIRCA-OS, SPAM, GAEZ+2015)",
               "All 252 five-map subsets: minimum",
               "All 252 five-map subsets: median",
               "All 252 five-map subsets: maximum"),
  existential_pct = round(100 * c(census_ev, min(ev5), median(ev5), max(ev5))),
  note = c(sprintf("percentile %.0f%%", 100 * mean(ev5 <= census_ev)),
           "", "", ""))
```

```
##                                                   quantity existential_pct
##                                                     <char>           <num>
## 1: Census-5 (GMIA, MIRCA2000, MIRCA-OS, SPAM, GAEZ+2015)              24
## 2:                       All 252 five-map subsets: minimum              24
## 3:                        All 252 five-map subsets: median              80
## 4:                       All 252 five-map subsets: maximum              90
##             note
##           <char>
## 1: percentile 0%
## 2:
## 3:
## 4:
```

3.2 R2.2: remote-sensing and census products are not a subset of one another

If remote sensing simply mapped a superset of census irrigation, it would never record zero where a census product detects irrigation. We test both directions and the disagreement within each family.

```
any_pos <- function(cols) rowSums(as.matrix(wide[, ..cols]) > 0, na.rm = TRUE) > 0
cP <- any_pos(census)
rP <- any_pos(rs)

data.table(
  quantity = c("RS records zero where census detects irrigation (% of census-irrigated cells)"
               "Census records zero where RS detects irrigation (% of RS-irrigated cells)",
               "Within RS family: GIAM vs GRIPC existential disagreement (%)",
               "Within census family: five FAO products existential disagreement (%)",
               "Mean global total, census maps (Mha)",
               "Mean global total, RS maps (Mha)"),
```

```r
  value = c(round(100 * sum(cP & !rP) / sum(cP)),
            round(100 * sum(rP & !cP) / sum(rP)),
            round(100 * exist_union(wide, rs)),
            round(100 * exist_union(wide, census)),
            round(mean(sapply(census, function(m) sum(wide[[m]])))),
            round(mean(sapply(rs,     function(m) sum(wide[[m]])))))
```

```
##                                                                              quantity
##                                                                                <char>
## 1: RS records zero where census detects irrigation (% of census-irrigated cells)
## 2:     Census records zero where RS detects irrigation (% of RS-irrigated cells)
## 3:                  Within RS family: GIAM vs GRIPC existential disagreement (%)
## 4:          Within census family: five FAO products existential disagreement (%)
## 5:                                           Mean global total, census maps (Mha)
## 6:                                               Mean global total, RS maps (Mha)
##    value
##    <num>
## 1:    36
## 2:    17
## 3:    57
## 4:    24
## 5:   244
## 6:   321
```

```r
# Per-map global totals underlying the census/RS means (which maps each group
# uses follows the reviewer's own classification in his comment) ---------------

data.table(
  group = c(rep("census", length(census)), rep("remote sensing", length(rs))),
  map = c(census, rs),
  total_mha = round(sapply(c(census, rs), function(m) sum(wide[[m]])), 1))
```

```
##             group       map total_mha
##            <char>    <char>     <num>
## 1:         census      gmia     245.8
## 2:         census mirca2000     210.5
## 3:         census  mirca_os     265.5
## 4:         census      spam     219.6
## 5:         census   gaez_v4     280.3
## 6: remote sensing      giam     335.2
## 7: remote sensing     gripc     306.0
```

3.3 R2.3: census products are compared on a common physical extent

The reviewer notes that the census products map different quantities (area equipped, harvested area, monthly growing area). We do not compare these raw quantities: in `code_original_datasets` we reduce every product to its actual physical irrigated extent, so these magnitude differences do not enter the comparison. The variable used for each product is recorded below.

```r
provenance <- data.table(
  product = c("GMIA", "SPAM2010", "MIRCA2000", "MIRCA-OS", "GAEZ+2015", "GIAM"),
  variable_used = c("Actually irrigated area (AEI x AAI fraction), not AEI",
                    "Physical irrigated area (SPAM 'A'), capped at cell area",
                    "Maximum monthly irrigated area (physical extent), not summed",
                    "Maximum monthly growing area (physical extent), not summed",
                    "Harvested irrigated area, capped at physical cell area",
                    "Net actually irrigated area (TAAI), not annualized (AIA)"),
  source = c("code_original_datasets.Rmd L254", "L439, L452", "L341", "L643",
             "L382, L399", "L205"))

knitr::kable(provenance, caption = "Variable extracted for each product (code_original_datasets
```

Table 2: Variable extracted for each product (code_original_datasets.Rmd). Every product is reduced to its actual physical irrigated extent, so differences between area equipped, harvested area and monthly growing area do not enter the existence comparison.

product	variable_used	source
GMIA	Actually irrigated area (AEI x AAI fraction), not AEI	code_original_datasets.Rmd L254
SPAM2010	Physical irrigated area (SPAM 'A'), capped at cell area	L439, L452
MIRCA2000	Maximum monthly irrigated area (physical extent), not summed	L341
MIRCA-OS	Maximum monthly growing area (physical extent), not summed	L643
GAEZ+2015	Harvested irrigated area, capped at physical cell area	L382, L399
GIAM	Net actually irrigated area (TAAI), not annualized (AIA)	L205

4 Reviewer 3

4.1 R3.2: where the maps agree on the existence of irrigation

The reviewer asks where all datasets agree on irrigation presence. We identify cells where all ten maps detect irrigation above the detectability threshold (1% of the cell, 500 ha at 0.2 deg) – the "agreement on presence" of the tau_max framework – and characterise them, including their overlap with rice production (EarthStat).

```
Apres <- as.matrix(wide[, ..maps])
wide[, agree_presence := rowSums(Apres > zero_tol_02 * 1e-6) == 10]

# Overlap with rice production (EarthStat, regridded to 0.2 deg) ---------------

rice <- rast(here("datasets", "earthstat_regridded", "res_02",
                  "rice_Production_ll_02.tif"))
wide[, rice_pos := { v <- terra::extract(rice, cbind(lon, lat))[, 1]
                     !is.na(v) & v > 0 }]

ag <- wide[agree_presence == TRUE]

data.table(
  metric = c("Agreement cells (all 10 above 1% of cell)",
             "In Asia (%)",
             "Latitude belt (10th-90th percentile)",
             "Coincide with rice production (%)",
             "All cropland cells coinciding with rice (%)"),
  value = c(as.character(nrow(ag)),
            as.character(round(100 * mean(ag$continent == "Asia"))),
            sprintf("%.0f-%.0f N", quantile(ag$lat, .10), quantile(ag$lat, .90)),
            as.character(round(100 * mean(ag$rice_pos))),
            as.character(round(100 * mean(wide$rice_pos)))))
```

```
##                                         metric   value
##                                         <char>  <char>
## 1:   Agreement cells (all 10 above 1% of cell)    9248
## 2:                                 In Asia (%)      79
## 3:        Latitude belt (10th-90th percentile) 18-42 N
## 4:           Coincide with rice production (%)      86
## 5: All cropland cells coinciding with rice (%)      56
```

```
knitr::kable(ag[, .N, country][order(-N)][1:5],
             caption = "Countries with the most cells where all ten datasets agree irrigation i
```

Table 3: Countries with the most cells where all ten datasets agree irrigation is present (0.2 deg).

country	N
China	2988
India	1921

country	N
United States	821
Pakistan	376
Iran	229

4.2 R3.14: disagreement does not track irrigation practice

The reviewer asks whether irrigation technique (e.g. drip in Spain) drives the disagreement. We report existential and extreme disagreement per country.

```
# Global reference ------------------------------------------------------------

glob <- data.table(country = "GLOBAL", cells = nrow(wide),
                   existential_pct = round(100 * exist_union(wide, maps)),
                   extreme_pct = round(100 * extreme_frac(wide, maps)))

# By country ------------------------------------------------------------------

ctys <- c("India", "Portugal", "Italy", "Pakistan", "China", "Spain",
          "Egypt", "France", "Mexico", "Iran", "United States")

res <- rbindlist(lapply(ctys, function(cn) {
  s <- wide[country == cn]
  if (nrow(s) == 0) return(NULL)
  data.table(country = cn, cells = nrow(s),
             existential_pct = round(100 * exist_union(s, maps)),
             extreme_pct = round(100 * extreme_frac(s, maps)))
}))

res <- rbind(glob, res[order(-extreme_pct)])

knitr::kable(res, caption = "Existential and extreme disagreement by country (0.2 deg). Existen
```

Table 4: Existential and extreme disagreement by country (0.2 deg). Existential: of the cells where at least one map detects irrigation, the percentage in which at least one map detects none. Extreme: percentage of cropland cells whose tau_max exceeds ten times the detectability threshold. Spain (the highest-drip country) is not a maximum.

country	cells	existential_pct	extreme_pct
GLOBAL	129628	90	23
India	6517	67	54
Portugal	236	88	49
Italy	819	76	42
Pakistan	1096	62	41
China	12317	71	40
Spain	1336	84	31
Egypt	304	69	31
France	1605	95	29

country	cells	existential_pct	extreme_pct
Iran	2931	88	29
Mexico	3126	92	25
United States	17087	93	15

5 Additional analyses: the non-identifiability claim

The three analyses below harden the claim that irrigation presence is non-identifiable (manuscript, section "Irrigated areas are non-identifiable"; Reviewer 1, comment 4; Reviewer 2, overall concern). The first counts how many complete global presence/absence fields are consistent with all ten maps at once. The second gives the rival hypothesis its best shot: if irrigation presence were an identifiable binary variable observed by ten imperfect instruments, the standard latent-class machinery for diagnostic tests without a gold standard (Hui and Walter 1980, Biometrics 36:167-171) should reproduce the observed response patterns with plausible error rates. The third tests the strongest surviving objection ("better data will fix it") against the best-in-class regional irrigation products over the United States.

5.1 Underdetermination: how many truths are consistent with the data

A cell is determined only if every map forces the same answer under every admissible detection convention $\tau \in [0, T]$: determined-present if the smallest reported area exceeds the strictest admissible threshold ($\min_d A_d > T$) and determined-absent if every map reports exactly zero. In all other cells both "irrigated" and "non-irrigated" are consistent with the ensemble, so the set of complete global presence fields consistent with all ten maps contains $2^{N_{\text{und}}}$ members. The universe here contains only cropland cells with at least one positive report, so determined-absent is zero by construction; adding all-zero cropland cells changes nothing because they are all determined-absent.

```
# UNDERDETERMINATION COUNT ########################################################

A <- as.matrix(wide[, ..maps])
Amin <- do.call(pmin, as.data.frame(A))
Amax <- do.call(pmax, as.data.frame(A))

underdet_fun <- function(T_ha) {
  T_mha    <- T_ha * 1e-6
  det_pres <- Amin > T_mha
  und      <- !(det_pres | Amax == 0)
  data.table(T_ha                = T_ha,
             det_present         = sum(det_pres),
             underdetermined     = sum(und),
             underdet_pct        = round(100 * mean(und), 1),
             existential_pct     = round(100 * sum(und & Amin == 0) / sum(und), 1),
             log10_truths        = round(sum(und) * log10(2)))
}

knitr::kable(rbindlist(lapply(c(100, 500, 1000), underdet_fun)),
             caption = "Underdetermination of irrigation presence (0.2 deg,
             cropland). det\\_present: cells where every map exceeds the
             strictest admissible threshold T. existential\\_pct: share of
             underdetermined cells where at least one map reports exactly
             zero. log10\\_truths: size (in powers of ten) of the set of
             complete global presence fields consistent with all ten maps.")
```

Table 5: Underdetermination of irrigation presence (0.2 deg, cropland). det_present: cells where every map exceeds the strictest admissible threshold T. existential_pct: share of underdetermined cells where at least one map reports exactly zero. log10_truths: size (in powers of ten) of the set of complete global presence fields consistent with all ten maps.

T_ha	det_present	underdetermined	underdet_pct	existential_pct	log10_truths
100	11147	118481	91.4	99.0	35666
500	9248	120380	92.9	97.4	36238
1000	7919	121709	93.9	96.4	36638

At the paper's threshold (T = 500 ha, 1% of the cell) 92.9% of the cells are underdetermined, so the ten maps are jointly consistent with $\approx 10^{36,238}$ distinct global irrigation-presence fields. Existential cases (at least one map reports exactly zero) make up 96-99% of the underdetermination across the admissible range, so the result is not driven by threshold sensitivity. Consistency checks: determined-present at T = 500 ha (9,248 cells) equals the R3.2 agreement-cell count, and the existential set (117,294 cells, 90.5%) equals the manuscript's 90% headline.

5.2 Latent-class analysis: testing "identifiable truth plus noise"

We fit K-class latent class models by expectation-maximisation on the 2^{10} response patterns of the binary presence matrix (conditional independence within class; 50 random starts per fit). Under the rival hypothesis, the two-class model (irrigated / non-irrigated plus per-map error rates) should fit. Three quantities matter: the goodness of fit of the two-class model, the per-map error rates it implies, and the structure of what it misses.

```
# K-CLASS LATENT CLASS MODEL VIA EM ON AGGREGATED RESPONSE PATTERNS ###########

lca_em <- function(pat, cnt, K, n_start = 50, tol = 1e-8, maxit = 3000,
                   seed = 123) {
  set.seed(seed)
  D <- ncol(pat); n <- sum(cnt)
  best <- NULL; logliks <- numeric(n_start)
  for (s in seq_len(n_start)) {
    x    <- rexp(K); pi_k <- x / sum(x)
    rho  <- matrix(runif(K * D, .05, .95), K, D)
    ll_old <- -Inf
    for (it in seq_len(maxit)) {
      logp <- sapply(seq_len(K), function(k)
        pat %*% log(rho[k, ]) + (1 - pat) %*% log(1 - rho[k, ]) + log(pi_k[k]))
      m    <- apply(logp, 1, max)
      lse  <- m + log(rowSums(exp(logp - m)))
      ll   <- sum(cnt * lse)
      post <- exp(logp - lse)
      w    <- post * cnt
      Nk   <- colSums(w)
      pi_k <- Nk / n
      rho  <- pmin(pmax(crossprod(w, pat) / Nk, 1e-6), 1 - 1e-6)
      if (abs(ll - ll_old) < tol) break
```

```r
      ll_old <- ll
    }
    logliks[s] <- ll
    if (is.null(best) || ll > best$ll)
      best <- list(ll = ll, pi = pi_k, rho = rho, post = post)
  }
  npar <- (K - 1) + K * D
  c(best, list(npar = npar, bic = -2 * best$ll + npar * log(n),
               logliks = logliks))
}

# Fit K = 1-4, report fit, error rates, residual dependence, 3-class profile --

lca_report_fun <- function(Y, label) {
  D    <- ncol(Y); n <- nrow(Y)
  code <- as.integer(Y %*% 2^(0:(D - 1)))
  tab  <- data.table(code = code)[, .N, by = code][order(code)]
  pat  <- t(sapply(tab$code, function(z) as.integer(intToBits(z)[1:D])))
  cnt  <- tab$N
  fits <- lapply(1:4, function(K) lca_em(pat, cnt, K))

  comp <- data.table(K = 1:4,
                     loglik = sapply(fits, function(f) round(f$ll, 1)),
                     npar   = sapply(fits, function(f) f$npar),
                     BIC    = sapply(fits, function(f) round(f$bic, 1)))

  f2  <- fits[[2]]
  irr <- which.max(rowMeans(f2$rho)); no <- setdiff(1:2, irr)
  err <- data.table(map = maps,
                    FNR = round(1 - f2$rho[irr, ], 3),
                    FPR = round(f2$rho[no, ], 3))[order(-FNR)]

  logp <- sapply(1:2, function(k)
    pat %*% log(f2$rho[k, ]) + (1 - pat) %*% log(1 - f2$rho[k, ]) +
      log(f2$pi[k]))
  m  <- apply(logp, 1, max)
  e  <- n * exp(m + log(rowSums(exp(logp - m))))
  G2 <- 2 * sum(cnt * log(cnt / e))
  df <- nrow(pat) - 1 - f2$npar

  P_obs <- crossprod(Y) / n
  P_exp <- f2$pi[1] * tcrossprod(f2$rho[1, ]) +
           f2$pi[2] * tcrossprod(f2$rho[2, ])
  R  <- (P_obs - P_exp) * 100
  ut <- which(upper.tri(R), arr.ind = TRUE)
  res <- data.table(m1 = maps[ut[, 1]], m2 = maps[ut[, 2]],
                    resid = round(R[ut], 2))[order(-abs(resid))][1:5]
```

```r
  f3  <- fits[[3]]
  ord <- order(rowMeans(f3$rho))
  pr  <- data.table(class = sprintf("pi = %.2f", f3$pi[ord]),
                    round(as.data.table(f3$rho[ord, , drop = FALSE]), 2))
  setnames(pr, c("class", maps))
  post3 <- f3$post[match(code, tab$code), ]
  mid_share <- 100 * mean(apply(post3, 1, which.max) == ord[2])
  ll_r  <- round(f2$logliks, 1)

  cat("\n####", label, "----", n, "cells,", nrow(pat), "patterns\n")
  print(comp)
  cat(sprintf("\n2-class fit: G2 = %.0f on df = %d (G2/df = %.1f)\n",
              G2, df, G2 / df))
  cat(sprintf("2-class multistart: %d distinct loglik modes in 50 starts\n",
              length(unique(ll_r))))
  cat("\nImplied per-map error rates (2-class):\n"); print(err)
  cat("\nLargest residual pairwise associations (obs - exp P(both = 1), x100):\n")
  print(res)
  cat(sprintf("\n3-class model: BIC improvement over 2-class = %.0f;",
              fits[[2]]$bic - f3$bic))
  cat(sprintf(" middle-class posterior share = %.1f%%\n", mid_share))
  cat("3-class item-probability profiles (rows ordered low to high):\n")
  print(pr)
  invisible(fits)
}
```

```r
# RUN AT BOTH DETECTION CONVENTIONS ##############################################

fits_tau0   <- lca_report_fun(1L * (A > 0),     "tau = 0 (any positive area)")
```

```
##
## #### tau = 0 (any positive area) ---- 129628 cells, 462 patterns
##        K    loglik  npar     BIC
##    <int>     <num> <num>   <num>
## 1:     1 -746739.4    10 1493597
## 2:     2 -580325.3    21 1160898
## 3:     3 -522411.1    32 1045199
## 4:     4 -513810.4    43 1028127
##
## 2-class fit: G2 = 165999 on df = 440 (G2/df = 377.3)
## 2-class multistart: 1 distinct loglik modes in 50 starts
##
## Implied per-map error rates (2-class):
##           map   FNR   FPR
##        <char> <num> <num>
##  1:      luh2 0.802 0.041
##  2:      giam 0.546 0.381
```

```
##  3:   nagaraj 0.504 0.085
##  4:     gripc 0.480 0.528
##  5:     meier 0.329 0.187
##  6: mirca2000 0.113 0.246
##  7:      spam 0.088 0.036
##  8:  mirca_os 0.018 0.106
##  9:   gaez_v4 0.013 0.006
## 10:      gmia 0.004 0.139
##
## Largest residual pairwise associations (obs - exp P(both = 1), x100):
##        m1      m2 resid
##    <char>  <char> <num>
## 1:  meier nagaraj 11.55
## 2:  gripc nagaraj  9.65
## 3:  gripc   meier  7.29
## 4:   giam nagaraj  7.04
## 5:   giam   gripc  6.05
##
## 3-class model: BIC improvement over 2-class = 115699; middle-class posterior share = 43.3%
## 3-class item-probability profiles (rows ordered low to high):
##        class gaez_v4  giam  gmia gripc  luh2 meier mirca2000 mirca_os nagaraj
##       <char>   <num> <num> <num> <num> <num> <num>     <num>    <num>   <num>
## 1: pi = 0.17    0.00  0.41  0.03  0.57  0.04  0.20      0.20     0.00    0.09
## 2: pi = 0.44    0.93  0.25  0.99  0.24  0.02  0.35      0.82     0.97    0.10
## 3: pi = 0.39    1.00  0.67  1.00  0.82  0.38  0.99      0.94     0.99    0.91
##     spam
##    <num>
## 1:  0.04
## 2:  0.81
## 3:  0.97
```

```
fits_tau500 <- lca_report_fun(1L * (A > 5e-4), "tau = 500 ha (1% of cell)")
```

```
##
## #### tau = 500 ha (1% of cell) ---- 129628 cells, 854 patterns
##        K    loglik  npar       BIC
##    <int>     <num> <num>     <num>
## 1:     1 -802738.4    10 1605594.6
## 2:     2 -481300.7    21  962848.5
## 3:     3 -458060.3    32  916497.3
## 4:     4 -452557.5    43  905621.2
##
## 2-class fit: G2 = 68582 on df = 832 (G2/df = 82.4)
## 2-class multistart: 1 distinct loglik modes in 50 starts
##
## Implied per-map error rates (2-class):
##           map   FNR   FPR
##        <char> <num> <num>
```

```
##  1:      luh2 0.619 0.031
##  2:      giam 0.449 0.175
##  3:     gripc 0.398 0.131
##  4:      spam 0.223 0.015
##  5: mirca2000 0.182 0.040
##  6:   nagaraj 0.116 0.153
##  7:  mirca_os 0.089 0.032
##  8:   gaez_v4 0.076 0.045
##  9:      gmia 0.066 0.010
## 10:     meier 0.028 0.100
##
## Largest residual pairwise associations (obs - exp P(both = 1), x100):
##           m1        m2 resid
##       <char>    <char> <num>
## 1:      giam     gripc  4.68
## 2: mirca2000      spam  2.36
## 3:      giam mirca2000  2.08
## 4:     gripc   nagaraj  2.06
## 5:     gripc      luh2  2.05
##
## 3-class model: BIC improvement over 2-class = 46351; middle-class posterior share = 15.5%
## 3-class item-probability profiles (rows ordered low to high):
##        class gaez_v4  giam  gmia gripc  luh2 meier mirca2000 mirca_os nagaraj
##       <char>   <num> <num> <num> <num> <num> <num>     <num>    <num>   <num>
## 1: pi = 0.60    0.02  0.17  0.00  0.13  0.03  0.07      0.03     0.01    0.13
## 2: pi = 0.16    0.76  0.24  0.66  0.27  0.14  0.86      0.45     0.66    0.68
## 3: pi = 0.24    0.97  0.71  1.00  0.76  0.50  1.00      0.97     0.99    0.96
##     spam
##    <num>
## 1:  0.01
## 2:  0.35
## 3:  0.96
```

The two-class model is decisively rejected at both conventions ($G^2/\text{df} = 377$ at $\tau = 0$ and 82 at $\tau = 500$ ha), and it is not rescued by estimation problems: all 50 starts converge to a single mode. Rescuing the rival hypothesis requires implausible instruments (at $\tau = 500$ ha, LUH2 missing 62% of truly irrigated cells, GIAM 45%, GRIPC 40%; at $\tau = 0$, GRIPC false-alarming in 53% of truly non-irrigated cells). The residual dependence concentrates within mapping families (GIAM-GRIPC on the remote-sensing side, MIRCA2000-SPAM on the census side), so the maps are not independent witnesses, and the third latent class that the data demand (BIC improvement of 46,000-116,000) is a census-positive / remote-sensing-negative class: the latent structure required by the data encodes how the maps were made, not only whether irrigation exists.

5.3 The best-products test over the United States

If existential disagreement reflected data quality alone, the best available irrigation products should agree on where irrigation exists. We test this in the country with the best irrigation data: we compare LANID (Xie and Lark 2021; 30 m, Landsat, annual), MIrAD-US (Pervez and Brown 2010;

250 m, MODIS anchored to USDA county statistics; version 4) and AIM-HPA (Deines et al. 2019; 30 m, Landsat, High Plains Aquifer only) in the two years all three cover (2012 and 2017). Decision rule, stated before computing: existential disagreement below 5% of union cells supports the claim that better data restore identifiability; disagreement comparable to the ten-map global ensemble over the same cells refutes it.

Each product was downloaded from its public archive (LANID: doi:10.5281/zenodo.5548555; MIrAD-US: doi:10.5066/P9NA3EO8; AIM-HPA: doi:10.4211/hs.a371fd69d41b4232806d81e17fe4efcb) and aggregated from its native equal-area grid to the paper's 0.2° grid by warp-summing the binary irrigated pixels and multiplying by the pixel area (0.09 ha at 30 m, 6.25 ha at 250 m), yielding irrigated hectares per cell. The chunk below documents the aggregation; it is not evaluated because it requires about 700 MB of raw rasters and several hours of warping. The aggregated table ships with the repository (`us_regional_products/`). As a sanity check, the CONUS totals are mutually consistent: LANID 23.1 (2012) and 24.2 Mha (2017); MIrAD 22.5 and 23.8 Mha.

```
# AGGREGATION OF THE US REGIONAL PRODUCTS (documentation; not run) ############

template <- rast(xmin = -126, xmax = -66, ymin = 24, ymax = 50,
                 resolution = 0.2, crs = "EPSG:4326")

agg_to_ha <- function(f, irr_val, px_ha) {
  ir <- (rast(f) == irr_val) * 1.0                # binary irrigated
  project(ir, template, method = "sum") * px_ha   # pixel count x pixel area
}

# lanid2012/2017.tif: 30 m, CONUS Albers, 0/1 (Zenodo record 5548555)
# mirad250_12/17v4.tif: 250 m, Lambert azimuthal EA, 0/1 (ScienceBase)
# {2012,2017}_AIM-HPA_rse_finalMapBinary.tif: 30 m; 0 = NoData, 1 = irrigated,
#   2 = not irrigated (HydroShare); validity = fraction of cell with value > 0
```

```
# EXISTENTIAL DISAGREEMENT: REGIONAL PRODUCTS VS THE GLOBAL ENSEMBLE ##########

# The CSV holds the product values extracted at the paper's 0.2 deg cell
# centres for every US cropland cell, so it joins back on lon/lat exactly
# (rounded to one decimal to be robust to floating-point noise).

reg <- fread(here("us_regional_products", "us_regional_products_02.csv"))
reg[, `:=`(lon = round(lon, 1), lat = round(lat, 1))]

us <- copy(wide[country == "United States"])
us[, `:=`(lon = round(lon, 1), lat = round(lat, 1))]
us <- merge(us, reg, by = c("lon", "lat"))
stopifnot(nrow(us) == nrow(reg))

# Existential: of union cells (>0 somewhere), % with at least one exact zero.
# Strong: % of union cells where one product is 0 and another exceeds 1% of
# the cell. Regional columns are in ha; the global maps are in Mha.

exist_pct <- function(d, cols) {
```

```r
  A  <- as.matrix(d[, ..cols])
  np <- rowSums(A > 0); un <- np > 0
  round(100 * sum(un & np < length(cols)) / sum(un), 1)
}
strong_pct <- function(d, cols, big) {
  A  <- as.matrix(d[, ..cols])
  un <- rowSums(A > 0) > 0
  round(100 * sum(un & apply(A, 1, min) == 0 &
                  apply(A, 1, max) > big) / sum(un), 1)
}
dir_pct <- function(d, a, b) {           # % of union cells with a = 0, b > 0
  A  <- as.matrix(d[, c(a, b), with = FALSE])
  un <- rowSums(A > 0) > 0
  round(100 * sum(un & A[, 1] == 0 & A[, 2] > 0) / sum(un), 1)
}

hpa <- us[aim_valid > 0.5]

row_fun <- function(d, universe, comparison, cols, big)
  data.table(universe, comparison,
             existential = exist_pct(d, cols),
             strong      = strong_pct(d, cols, big))

out <- rbind(
  row_fun(us,  "CONUS",       "LANID x MIrAD 2012", c("lanid12", "mirad12"), 500),
  row_fun(us,  "CONUS",       "LANID x MIrAD 2017", c("lanid17", "mirad17"), 500),
  row_fun(us,  "CONUS",       "Ten global maps",    maps,                    5e-4),
  row_fun(hpa, "High Plains", "LANID x MIrAD 2012", c("lanid12", "mirad12"), 500),
  row_fun(hpa, "High Plains", "LANID x MIrAD 2017", c("lanid17", "mirad17"), 500),
  row_fun(hpa, "High Plains", "LANID x MIrAD x AIM 2012",
          c("lanid12", "mirad12", "aim12"), 500),
  row_fun(hpa, "High Plains", "LANID x MIrAD x AIM 2017",
          c("lanid17", "mirad17", "aim17"), 500),
  row_fun(hpa, "High Plains", "Ten global maps",    maps,                    5e-4))

knitr::kable(out, caption = "Existential disagreement (of the cells where at
             least one product records irrigation, the percentage where at
             least one records exactly zero) and strong contradictions (one
             product records zero, another exceeds 1\\% of the cell) among the
             best US regional irrigation products, benchmarked against the ten
             global maps over the same cells. CONUS: 17,087 US cropland cells;
             High Plains: 1,549 cells within the AIM-HPA footprint.")
```

Table 6: Existential disagreement (of the cells where at least one product records irrigation, the percentage where at least one records exactly zero) and strong contradictions (one product records zero, another exceeds 1% of the cell) among the best US regional irrigation products, benchmarked against the ten global maps over the same cells. CONUS: 17,087 US cropland cells; High Plains: 1,549 cells within the AIM-HPA footprint.

universe	comparison	existential	strong
CONUS	LANID x MIrAD 2012	32.1	1.3
CONUS	LANID x MIrAD 2017	33.6	3.0
CONUS	Ten global maps	93.0	46.6
High Plains	LANID x MIrAD 2012	5.7	0.0
High Plains	LANID x MIrAD 2017	5.3	0.2
High Plains	LANID x MIrAD x AIM 2012	7.8	0.3
High Plains	LANID x MIrAD x AIM 2017	8.0	0.7
High Plains	Ten global maps	75.0	70.7

```
# Directional asymmetry (mirrors the global census-vs-remote-sensing axis) -----

data.table(
  year = c(2012, 2017),
  lanid0_mirad_pos = c(dir_pct(us, "lanid12", "mirad12"),
                       dir_pct(us, "lanid17", "mirad17")),
  mirad0_lanid_pos = c(dir_pct(us, "mirad12", "lanid12"),
                       dir_pct(us, "mirad17", "lanid17")))
```

```
##     year lanid0_mirad_pos mirad0_lanid_pos
##    <num>            <num>            <num>
## 1:  2012             23.1              9.0
## 2:  2017             26.0              7.6
```

The verdict is split and informative. Across the CONUS, the two best national products, in the same years and anchored to US ground data, disagree on the existence of irrigation in a third of the cells where either sees it, decisively failing the 5% bar; the direction mirrors the global census-versus-remote-sensing axis (the USDA-anchored MIrAD sees irrigation where the Landsat-based LANID sees none in 23-26% of union cells). In the High Plains Aquifer, the best-mapped, centre-pivot, high-contrast region, the products nearly converge (5-8%). Strong contradictions almost vanish everywhere (0-3% against 47-70% for the global ensemble). Better data thus fix magnitude-scale contradictions and resolve the clear cases, but leave the low-intensity middle contested: exactly the structure predicted if "irrigated" admits clear cases at the extremes and no principled boundary in the middle, and against the expectation that more resolution and better calibration restore identifiability in general.

6 Session information

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.3.1
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framewo
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
##
## locale:
## [1] C
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
## [1] magrittr_2.0.5    terra_1.8-93      here_1.0.2        data.table_1.18.4
##
## loaded via a namespace (and not attached):
##  [1] vctrs_0.7.3        cli_3.6.6          knitr_1.51         rlang_1.2.0
##  [5] xfun_0.55          otel_0.2.0         generics_0.1.4     S7_0.2.2
##  [9] glue_1.8.1         rprojroot_2.1.1    htmltools_0.5.9    scales_1.4.0
## [13] sensobol_1.2.0     rmarkdown_2.30     grid_4.5.2         tibble_3.3.1
## [17] evaluate_1.0.5     fastmap_1.2.0      yaml_2.3.12        lifecycle_1.0.5
## [21] compiler_4.5.2     codetools_0.2-20   dplyr_1.1.4        RColorBrewer_1.1-3
## [25] Rcpp_1.1.1-1.1     pkgconfig_2.0.3    farver_2.1.2       digest_0.6.39
## [29] R6_2.6.1           tidyselect_1.2.1   Rdpack_2.6.6       pillar_1.11.1
## [33] rbibutils_2.4.1    tools_4.5.2        gtable_0.3.6       ggplot2_4.0.3.9000
```